Reference-country composition changes the interpretation of PCOS-attributed infertility comparisons: GBD 2023 and UN WPP 2024

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Running title: Infertility comparator composition

Extended abstract

Study question: Does the population composition of a pooled international comparator materially change the interpretation of modeled PCOS-attributed infertility rates for China, even after age standardization?

Summary answer: Demographic change in the 18-country comparator accounted for a substantial part of the narrowing modeled PCOS contrast; using independent United Nations population weights yielded a similar accounting result, while a near-zero secondary-infertility comparison reversed sign under other reasonable benchmark choices.

What is known already: Previous GBD analyses have described PCOS-attributed and overall infertility burden, primary and secondary subtypes, trends and forecasts. A pooled reference group's country shares may change within age groups even when the age-standardization weights are fixed; this changes the comparison being estimated.

Study design, size, duration: Descriptive secondary analysis of GBD 2023, 1990–2023, for women aged 15–49 years in China and the other 18 sovereign G20 countries. A separate population-source sensitivity used historical 1990–2023 UN World Population Prospects 2024

midyear population counts for 19 countries, 34 years and seven five-year age groups (4,522 country–year–age cells).

Participants/materials, setting, methods: GBD 2023 cause-specific infertility prevalence (PCOS primary, secondary and total; endometriosis as a contrasting cause) was age-standardized within ages 15–49. The annual country-pooled comparator was decomposed symmetrically into changes in Chinese modeled rates, the 18 reference-country modeled rates and their population shares within age. In an independent demographic-source audit, original age-specific GBD modeled disease rates were held unchanged while the reference-country weights were replaced with corresponding UN WPP 2024 population shares. Alternative fixed-1990 and equal-country benchmarks, end-year changes, age restriction and country exclusions assessed choice sensitivity.

Main results and the role of chance: The China-minus-reference PCOS-related total-infertility difference changed by +151.54 per 100,000 women from 1990 to 2023; a symmetric comparator-share component was +88.65 (58.5% of the signed net change) under GBD population shares. Reweighting the reference with UN population produced a net change of +150.69 and share component +86.76 (57.6%). The 2023 PCOS secondary-infertility difference was +2.26 per 100,000 under UN contemporary country shares, but -48.34 under fixed-1990 shares. These are deterministic model-conditioned contrasts; no jointly indexed GBD posterior draws were available for derived uncertainty intervals.

Limitations, reasons for caution: UN WPP independently estimates demographic weights, not disease burden, and may share upstream demographic information with GBD. The comparator is an economically selected set of countries, not a clinical control or the world. Standard age weights were reconstructed from rounded published values. GBD causes are modeled and not person-level clinical diagnoses; endometriosis 2023 estimates were endpoint-sensitive. Analyses were expanded post hoc, and full GBD cross-cell posterior uncertainty was unavailable.

Wider implications of the findings: International comparisons should specify the weighting estimand and assess plausible comparator compositions before interpreting numerical convergence or subgroup rank changes as epidemiological progress. These findings do not establish causal risk, improved infertility care or national treatment effects.

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Key words: infertility; polycystic ovary syndrome; population composition; age standardization; GBD 2023.

1 Introduction

Infertility associated with polycystic ovarian syndrome (PCOS) and endometriosis is relevant to population reproductive health as well as individualized clinical care Teede et al. (2023); Becker et al. (2022). Estimates attributed to these causes within the Global Burden of Disease (GBD) infertility impairment describe a different outcome from total PCOS or endometriosis prevalence. They are also distinct from lifetime infertility or infertility measured exclusively among people currently trying to conceive GBD 2023 Collaborators (2025).

Previous studies have already quantified global PCOS-attributed infertility through GBD 2019 Liu et al. (2024), described global infertility trends and forecasts in this journal using GBD 2021 Liang et al. (2025), and examined endometriosis, PCOS and primary/secondary infertility attribution with GBD 2021 Xu et al. (2025). China–global comparisons Lu et al. (2026), GBD 2021/2023 analyses Wang et al. (2026) and general GBD trend/decomposition studies Feng et al. (2025) further constrain any novelty claim. Merely updating the GBD version, adding countries, reporting subtypes or forecasting future rates would duplicate existing analytical territory. We therefore investigate a different estimand question: the changing demographic composition of a pooled international comparison group.

An unresolved interpretive question concerns the interpretation of an aggregate international contrast. A population-pooled reference combines country rates with country population shares that may change over time, even after age standardization. A narrowing difference could therefore reflect changing Chinese rates, changing reference-country rates, or changing reference-country composition. Primary and secondary infertility may contribute differently to each component. These are separable accounting questions; they do not identify causal mechanisms.

We compared China's cause-specific infertility estimates with a fixed set of 18 other sovereign G20 countries. We quantified subtype and age contributions to changes in the contrast and added an explicit decomposition of reference composition during manuscript review. We also assessed country and endpoint dependence. Endometriosis was retained as a contrasting cause,

with terminal-year instability treated as a finding requiring caution rather than evidence of sustained convergence.

2 Materials and methods

2.1 Design, data and outcome definitions

This ecological secondary analysis used GBD 2023 Results Tool downloads obtained on 25 October 2026 IHME (2026). The analysis included females aged 15–49, seven five-year age groups and annual estimates for 1990–2023. China was compared with Argentina, Australia, Brazil, Canada, France, Germany, India, Indonesia, Italy, Japan, Republic of Korea, Mexico, Russian Federation, Saudi Arabia, South Africa, Türkiye, United Kingdom and United States of America. These are the other sovereign country members of the G20; regional organizations were excluded. The G20 is an economically defined comparison, not a matched control population or a global benchmark.

We selected infertility impairment 198, underlying causes endometriosis (607) and PCOS (605), prevalence and years lived with disability (YLDs), and Number and Rate. Primary (220) and secondary (221) infertility were downloaded separately. Matched female populations came from the same release. Main analyses concern prevalence; YLDs, case-count decompositions, country distributions, trend models and failed forecast validation are secondary results in Additional file 1. The present study uses aggregate GBD model estimates and UN population counts only; no individual participant records were analysed for this manuscript.

The GBD methods appendix describes survey-based primary infertility in couples without a previous live birth who desire a child, have been in a union for at least five years, and report no contraceptive use. Secondary infertility also uses the interval since the last live birth. Heterogeneous survey and clinical definitions are harmonized within GBD; the resulting estimates should not be equated with a uniform 12-month clinical definition. Underlying-cause allocation occurs within the modeled infertility envelope GBD 2023 Collaborators (2025). The denominator here is all females in each age group, not only couples attempting pregnancy.

2.2 Validation and standardization

We required complete dimensional coverage, unique cells, consistent cause and infertility identifiers, finite non-negative values and ordered original marginal uncertainty bounds. Number divided by matching population was compared with official Rate. Annual subtype rates were reconstructed from Number and matched population; independent original Number/Rate exports for 1990 and 2023 checked this reconstruction. Primary plus secondary point estimates were required to reproduce the total. Input hashes, mappings and retrieval specifications accompany the code PCOS research workspace (2026).

We used published GBD 2023 standard-population weights reproduced to six decimals, normalized over ages 15–49 Chen et al. (2026). The original machine-precision IHME weight file was not retrieved. Thus, all standardized rates are restricted-age rates. With normalized age weight w_a, country c, age a, year t, infertility type k, modeled cases C and female population P, rates per 100,000 were $r_{catk} = 10^5 C_{catk}/P_{cat}$. The age-specific reference was formed before standardization:

$$r_{Ratk} = \sum_{c \in R} s_{cat} r_{catk}, \quad s_{cat} = \frac{P_{cat}}{\sum_{j \in R} P_{jat}}, \quad A_{Ctk} = \sum_a w_a r_{Catk}, \quad A_{Rtk} = \sum_a w_a r_{Ratk}. \tag{1}$$

Here R always excludes China. The country-share weights s vary within age and year; the standard age weights w are fixed. Rates were not constructed by averaging country standardized rates.

2.3 Subtype and reference-composition accounting

We calculated the signed difference $D_{tk} = A_{Ctk} - A_{Rtk}$ and ratio A_{Ctk}/A_{Rtk}. Because primary and secondary counts sum to total, their differences and changes also sum exactly. Their signed contributions are not attributable fractions or causal effects. Each age group's contribution is its standard weight multiplied by its age-specific difference change.

During manuscript review, we added a symmetric decomposition between start year 0 and end

year T:

$$\Delta D_k = X_k + Q_k + S_k, \tag{2}$$

$$X_k = \sum_a w_a (r_{CaTk} - r_{Ca0k}), \tag{3}$$

$$Q_k = -\sum_a w_a \sum_{c \in R} \frac{s_{ca0} + s_{caT}}{2} (r_{caTk} - r_{ca0k}), \tag{4}$$

$$S_k = -\sum_a w_a \sum_{c \in R} \frac{r_{ca0k} + r_{caTk}}{2} (s_{caT} - s_{ca0}). \tag{5}$$

X is the Chinese rate component, Q the reference-country rate component and S the reference country-share component. The mean of the two replacement orders shares the product interaction symmetrically. Components sum exactly to the change in the standardized contrast, separately by subtype and age. A positive component moves China minus reference upward; with an initially negative contrast, it can indicate narrowing without indicating health improvement. This decomposition describes reference weighting, whereas the separate case-count decomposition concerns China's own population size and age distribution.

2.4 Independent UN population benchmark audit

To test whether the reference-composition finding depended on using the GBD population series, we downloaded the original United Nations World Population Prospects (WPP) 2024 *PopulationByAge5GroupSex_Medium* CSV from the UN Population Division UN DESA (2024). Only the historical 1990–2023 mid-year (1 July) estimates were used; WPP population projections from 2024 onwards were excluded. We selected females, the same 19 sovereign G20 countries and seven five-year age groups, using ISO3 country identifiers. Each of the expected 4,522 country–year–age records was present once. Source values in thousands were converted to persons. Original compressed-file and filtered-subset hashes are reported with reproducibility scripts.

The WPP counterbenchmark substituted only the reference-country *shares within age and year*:

$$s_{cat}^{\mathrm{UN}} = \frac{P_{cat}^{\mathrm{UN}}}{\sum_{j \in R} P_{jat}^{\mathrm{UN}}}, \qquad A_{Rt}^{\mathrm{UN}} = \sum_a w_a \sum_{c \in R} s_{cat}^{\mathrm{UN}} r_{cat}^{\mathrm{GBD}}. \tag{6}$$

Here r_{cat}^{GBD} denotes the original GBD 2023 age-specific prevalence rate, retained *unchanged* under both population-source scenarios. The 15–49 standard age weights w_a likewise remained un-

changed. We did not recalculate disease incidence/prevalence rates using UN denominators, which would mix incompatible modeled cases and populations. Symmetric rate-versus-country-share decompositions, 2021–2023 endpoints, 1990 fixed-share comparisons, equal-country benchmarks and country exclusions were repeated with both population sources. Country-share discrepancies were quantified using total-variation distance, one half of the sum of absolute country-share differences within age and year. All analyses were exploratory, model-conditional and descriptive, without inferential uncertainty for derived GBD contrasts.

2.5 Sensitivity analyses and uncertainty

We excluded each reference country in turn, restricted ages to 20–49, and fixed within-age reference-country shares at their 1990 values. The symmetric decomposition was repeated for country exclusions, the restricted age range, periods starting in 1990 or 2010, and endpoints 2021, 2022 and 2023. Endpoint analyses were added after inspection of a terminal-year endometriosis increase; the reference-composition decomposition was added during manuscript review. Neither was externally preregistered. Fixed-share contrasts are a separate, order-specific sensitivity, not an interchangeable estimate of the symmetric share component.

Additional terminal-year and published-definition cross-check (post hoc). Following identification of the unusual final-year endometriosis trajectory, we repeated direct age standardization on the original GBD 2023 age-specific rates in all 19 countries, calculated 2022–2023 year-on-year changes and compared the China endpoint with its own 1991–2022 changes (a descriptive stress test, not a null-hypothesis test). Chinese primary and secondary changes were summed as standard-weighted within-age differences. In addition, we compared the original China modeled case count and published endometriosis rates from a separately authored 2026 publication based on the *same* GBD 2023 release Liu et al. (2026). To harmonize units, we contrasted the seven age weights used as printed (which total 0.494902 for ages 15–49) with the normalized restricted-age weights used throughout this manuscript. Publication-level agreement on a common source is not independent population-based validation, and the published clinical subset has no national denominator.

We examined all 128 combinations of perturbing each unnormalized standard weight by half of its last published decimal. These deterministic rounding envelopes do not measure epidemi-

ological uncertainty. Additional file 1 describes all secondary methods, including conditional trend intervals and forecast-validation rules.

Jointly aligned GBD draws across ages, countries and years were unavailable. We therefore report derived point estimates without propagated uncertainty intervals or significance tests. Original marginal limits are preserved but not summed or treated as standard errors. Country exclusions describe reference sensitivity, not sampling uncertainty. YLDs and prevalence share inputs and are not independent replications. Python 3.12 and pinned NumPy, pandas, SciPy, statsmodels and Matplotlib dependencies were used. Code and machine-readable outputs are available in Additional file 2 and the repository PCOS research workspace (2026). Applicable GATHER items, including incomplete propagated uncertainty reporting, are mapped in Additional file 1 Stevens et al. (2016).

2.6 Ethical considerations

This secondary analysis uses public, aggregate GBD estimates and UN population summaries without individually identifiable records. XX: the responsible investigator must verify whether institutional review or a documented exemption/non-human-subject determination is required. No institutional approval or waiver is asserted.

2.7 Use of artificial intelligence

OpenAI-assisted tools were used to support code development, document drafting, and technical checks in this project. No artificial observations were substituted for the downloaded GBD data. Final human review of code, sources, interpretation and all author declarations remains necessary before submission. The tool is not an author and cannot approve the manuscript.

3 Results

3.1 Coverage and overall contrasts

The total burden export contained 36,176 rows and the population export 4,522 age–country–year cells. A further 36,176 annual subtype Number rows and 4,256 endpoint Number/Rate rows covered both causes and measures. Primary plus secondary reproduced total point

estimates throughout. Reconstructed subtype endpoint rates agreed with original rates to within 1.51×10^{-11} per 100,000. These checks establish extraction and arithmetic consistency, not independent validity of GBD estimates.

China's PCOS-related infertility standardized prevalence increased from 333.56 to 605.08 per 100,000, compared with 606.48 to 726.45 in the reference (Table 1; Figure 1). The difference changed from −272.92 to −121.38 and the ratio from 0.550 to 0.833. For endometriosis, rates changed from 63.70 to 53.30 in China and 66.63 to 53.51 in the reference. The 2023 total difference was −0.20; it was −8.76 in 2022. China's endometriosis estimate rose by 23.6% in the last year, so the terminal comparison does not summarize a stable trajectory.

GBD 2023 terminal-year source audit. China's endometriosis-attributed infertility rate rose from 43.13 to 53.30 per 100,000 between 2022 and 2023 (+23.58%); the absolute 15–49 standardized increase was +10.17 per 100,000. Its primary and secondary components were +0.74 and +9.43 per 100,000, respectively, so the latter made up 92.7% of the *signed modeled rate increment*, not an etiologic fraction. Over the preceding modeled 1991–2022 intervals, China's largest positive annual change was +1.56%. The 2022–2023 values for endometriosis were +14.82% in the United States, +11.93% in Japan and −0.41% in the Republic of Korea, while China's PCOS-related infertility rose only +0.50% (Supplementary Figures S7 and S8). These heterogeneous model estimates cannot establish a new biological event in 2023.

Our 2023 China total of 181,587.09 modeled endometriosis-infertility cases agrees with the 181,587 cases in the separately published GBD 2023 analysis Liu et al. (2026). Their 1990 and 2023 reported age-standardized rates (31.52 and 26.38 per 100,000) closely matched our calculations using the *unrenormalized* seven age weights (31.524 and 26.380); our restricted-15–49 normalized figures were 63.697 and 53.304. The 0.494902 scaling factor explains this otherwise apparently discordant comparison; the two conventions should not be combined as though they were the same estimand. This shared-source numerical cross-check is not independent validation (Supplementary Table S8).

3.2 Primary and secondary infertility

In 2023, the PCOS primary-infertility difference was −123.54 per 100,000 (ratio 0.441), whereas the secondary difference was +2.16 (ratio 1.004; Table 2; Figure 2). Their changes since 1990 were

Table 1: Total infertility prevalence in China and the reference population.

		Cases, thousands		Rate per 100,000			
Cause	Year	China	Reference	China	Reference	*D*	Ratio
Endometriosis	1990	192.8	375.6	63.70	66.63	-2.93	0.956
Endometriosis	2023	181.6	457.1	53.30	53.51	-0.20	0.996
PCOS	1990	1083.3	3446.2	333.56	606.48	-272.92	0.550
PCOS	2023	1866.7	6124.2	605.08	726.45	-121.38	0.833

Rates standardized within ages 15–49. Reference pools the other 18 sovereign G20 countries within age. *D*: China minus reference. All cells are modeled point estimates without propagated uncertainty.

+46.40 and +105.14, respectively, summing to +151.54. Thus secondary infertility accounted for 69.4% of the signed total change. The share was 70.5% ending in 2021 and 69.0% ending in 2022. Across country exclusions and restriction to ages 20–49, the 2023 share ranged from 64.2% to 75.4%; fixing reference-country shares at 1990 yielded 78.1%. The exact percentage therefore depends on reference construction, while secondary infertility remained the larger positive component in these checks.

The 2023 PCOS primary difference remained negative under every country exclusion (-201.52 to -101.89), but the secondary difference ranged from -113.63 to +30.72. Near parity of secondary infertility was therefore a main-reference result, not a stable sign conclusion. For endometriosis, the 2023 primary difference of -7.94 and secondary difference of +7.73 nearly canceled. Both subtype signs survived country exclusions in 2023, but the secondary difference was -0.40 in 2022. Its endpoint dependence limits the interpretation of cancellation.

Table 2: Primary and secondary infertility prevalence at the endpoints.

Cause	Type	Year	China rate	Ref. rate	*D*	Ratio
Endometriosis	Primary	1990	4.92	19.64	-14.72	0.250
Endometriosis	Primary	2023	4.94	12.87	-7.94	0.384
Endometriosis	Secondary	1990	58.78	46.98	11.79	1.251
Endometriosis	Secondary	2023	48.37	40.63	7.73	1.190
PCOS	Primary	1990	44.70	214.64	-169.94	0.208
PCOS	Primary	2023	97.35	220.89	-123.54	0.441
PCOS	Secondary	1990	288.86	391.84	-102.98	0.737
PCOS	Secondary	2023	507.72	505.56	2.16	1.004

Rate units: per 100,000 females; standardization restricted to ages 15–49. Point estimates only. Annual rates were reconstructed from Number and same-release population, with original endpoint Rate checks. Types sum to total before displayed rounding.

3.3 Changing reference composition

For PCOS, Chinese rate changes contributed +271.51 per 100,000 to the changing contrast, reference-country rate changes −208.62, and reference-country share changes +88.65 (Table 3; Figure 3). An additional exploratory annual-chained decomposition gave a share component of +87.11 (57.5%), compared with the endpoint value +88.65 (58.5%). This is a deterministic time-path sensitivity, not uncertainty estimation. The endpoint share component accounted for 58.5% of the +151.54 signed net change; the combined rate components accounted for +62.89. The corresponding share components were +88.11 ending in 2021 and +86.74 ending in 2022. Across country exclusions and the age-range sensitivity, the 2023 composition share ranged from 39.8% to 67.9%. A material reference-composition contribution persisted, although a majority contribution did not hold in every specification.

For primary PCOS infertility, the three components were +52.65, −38.09 and +31.84; for secondary infertility, they were +218.86, −170.53 and +56.81. Thus subtype and reference composition are intersecting dimensions of the same contrast change: the 69.4% subtype share and 58.5% composition share must not be added. Fixing all reference-country shares at 1990 instead yielded a 2023 total difference of −202.48, compared with −121.38 under current pooling. This fixed-share calculation follows one replacement path and therefore differs from the symmetric component.

Endometriosis's +2.73 net difference change combined Chinese rate changes of −10.39, reference-country rate changes of +13.50 and reference composition of −0.38. Its near-zero net change makes percentage contributions unstable; we report absolute components. Age-by-type component tables for each endpoint accompany Additional file 2.

3.4 Independent UN population benchmark and comparator sensitivity

The independent UN WPP 2024 file matched the 19-country GBD study by year and five-year female age group, with complete 4,522-cell coverage. Its 2023 within-age reference-country shares differed from the GBD population shares by a mean total-variation distance of 1.32 percentage points across seven ages (mean 1.13 over 1990–2023). Under UN weighting, India's age-weighted share of the reference increased from 35.71% in 1990 to 46.45% in 2023, while Japan's declined from 5.65% to 2.82%. These are changes in reference composition, not changes in disease risk.

Table 3: Symmetric decomposition of changes in standardized differences, 1990–2023.

Cause	Type	Net change	China rates	Ref. rates	Ref. shares
Endometriosis	Primary	+6.79	+0.02	+6.69	+0.08
Endometriosis	Secondary	-4.06	-10.41	+6.81	-0.46
Endometriosis	Total	+2.73	-10.39	+13.50	-0.38
PCOS	Primary	+46.40	+52.65	-38.09	+31.84
PCOS	Secondary	+105.14	+218.86	-170.53	+56.81
PCOS	Total	+151.54	+271.51	-208.62	+88.65

All components use per 100,000 females and sum to net change. Reference shares are country population shares within each age. Positive values move China minus reference upward. Percentages of a signed net change are accounting summaries, not causal fractions.

For total PCOS-attributed infertility, the GBD-weighted China–reference contrast changed by +151.54 per 100,000 from 1990 to 2023, of which the symmetric country-composition term was +88.65 (58.5%). Replacing only the comparator shares with UN WPP population gave a contrast change of +150.69, and a country-composition contribution of +86.76 (57.6%; Table 4). The resulting 2023 contrast was -122.62 per 100,000 versus -121.38 with GBD population shares (Figure 4). The UN estimates thus corroborate the numerical stability of this accounting result to an alternative demographic source, not the validity of the underlying infertility rates.

The subtype comparison remained benchmark-dependent. With UN contemporary shares, the 2023 PCOS primary difference was -124.88 and the secondary difference was +2.26 per 100,000. With UN country shares fixed at 1990, the secondary difference became -48.34; using equal country shares it was -127.89. An apparently small secondary excess is therefore not a robust international ranking. The UN composition term for the total PCOS contrast remained positive at +86.50 with a 2021 endpoint and +85.20 with a 2022 endpoint. Its total PCOS difference remained negative in all 18 UN leave-one-reference-country-out analyses (range -318.33 to -82.26 per 100,000 in 2023).

The endometriosis total contrast was -0.20 per 100,000 with GBD country shares and -0.03 with UN population shares in 2023. Both were strongly dependent on using 2023 rather than 2022 as endpoint; the corresponding UN-weighted 2022 difference was -8.61. This population-source audit does not independently confirm an epidemiological endometriosis increase.

Table 4: Independent UN WPP 2024 population-source sensitivity of 1990–2023 GBD-derived infertility contrasts.

Cause	Shares	D_{1990}	D_{2023}	ΔD	Country share	Share/ΔD
PCOS	GBD	−272.92	−121.38	+151.54	+88.65	58.5%
PCOS	UN WPP	−273.31	−122.62	+150.69	+86.76	57.6%
Endometriosis	GBD	−2.93	−0.20	+2.73	−0.38	−13.8%
Endometriosis	UN WPP	−2.75	−0.03	+2.72	−0.38	−14.0%

All rates, differences and decomposition contributions are modeled point estimates per 100,000 women aged 15–49. UN changes only reference-country population shares within age; GBD prevalence rates and fixed standard age weights are unchanged. Percentage attribution for endometriosis is numerically unstable and not used for substantive interpretation. No draw-propagated uncertainty is available.

3.5 Age patterns and secondary analyses

Ages 35–39 and 40–44 contributed +31.07 and +33.60 per 100,000, respectively, to the PCOS difference change, together 42.7% of the total. These are standard-weighted contrast contributions, not case counts. The age-by-type table separates Chinese rates, reference rates and reference composition within each cell.

China's PCOS-related infertility count rose by 783,425. A separate case-count decomposition assigned +860,721 to age-specific prevalence, offset by $-68,426$ from female population size and $-8,869$ from age distribution within ages 15–49. This is consistent with increasing modeled prevalence despite demographic offsets; it does not identify its causes. The total PCOS difference remained negative across reference exclusions. The endometriosis total difference changed sign across country exclusions. Rounding the published standard weights within their evaluated precision did not reverse the main 2023 signs. Secondary YLD, trend and benchmark results are retained in Additional file 1. Forecast candidates failed the comparative validation rule, so no long-range forecasts are reported.

4 Discussion

The principal finding is that subtype composition and reference population composition both affect the interpretation of international infertility contrasts. China's negative PCOS-related infertility difference narrowed, but a substantial part of that change corresponded to population redistribution among reference countries. At the same time, secondary infertility accounted for most of the total contrast change, while a primary-infertility difference persisted. These results

concern how an international comparison is constructed and what its summary conceals. A separate, independently analysed US survey is not included in this focused manuscript because its diagnosis-based fertility outcomes differ from the GBD cause-attributed estimates.

This is an incremental contribution to an extensively studied field. Previous work already examined these causes, their primary and secondary infertility attribution, China/global patterns and standard trend and decomposition methods Xu et al. (2025); Lu et al. (2026); Wang et al. (2026); Feng et al. (2025). Our contribution is the joint accounting of subtype and within-age reference-country shares, with sensitivity to country inclusion and endpoint. We do not propose a new decomposition method or claim that GBD 2023 or subtype stratification alone is novel. The value is an interpretable application that changes the substantive reading of apparent convergence.

Age standardization removes changes in the chosen age weights, but it does not fix the mix of countries within an age group. When countries with different infertility rates gain or lose population share, a pooled reference can move even with unchanged country rates. The symmetric identity quantifies this component separately from the rate component and avoids assigning the entire rate–share interaction to one replacement order. The fixed-1990-share sensitivity gives a complementary contrast under a different reference definition. Neither is a preferred target rate or a causal counterfactual.

The subtype results further discourage describing narrowing as improved reproductive health. Both Chinese and reference PCOS rates increased. Secondary infertility's larger contribution does not show that individual women moved from primary to secondary infertility, nor that parity history caused the increase. The GBD categories reflect modeled estimates and survey definitions; diagnosis, fertility intentions, exposure to pregnancy, treatment and underlying disease attribution could all affect interpretation, but were not directly measured. The all-female denominator cannot estimate the risk among couples trying to conceive.

For endometriosis, the final-year near cancellation should be viewed as an unstable contrast. Independent download checks reproduced the source values, yet these checks cannot determine whether the abrupt terminal-year change reflects new evidence, model behavior or an actual epidemiological shift. Analyses ending in 2021 or 2022 gave a materially different overall picture. This unresolved source-level issue warrants investigation with independent evidence before a substantive convergence claim. Retaining the earlier endpoints and full annual trajectories

makes the limitation visible.

A further GBD 2023 study published in August 2026 reported the same modeled Chinese endometriosis-infertility total and a post-2021 rise, alongside a selected series of 168 clinically confirmed patients Liu et al. (2026). Matching its reported burden required identifying that its published rate scale corresponded numerically to weights for ages 15–49 *without* renormalizing them, whereas our estimand standardizes within the specified age range. Neither the common GBD model output nor the clinical case series can establish independent population prevalence or determine why the 2023 final-year modeled values changed. The large modeled increase in China, Japan and the United States but not the Republic of Korea suggests that input or modelling diagnostics by location are warranted, not that a single healthcare or pandemic mechanism has been demonstrated. The pooled WHO evidence on overall infertility cannot directly validate GBD's cause assignment WHO (2023). This terminal-year and scale cross-check was added after outcome inspection and therefore should be treated as exploratory.

Case-count accounting answers a different public-health question. The PCOS increase occurred despite offsets from China's reproductive-age female population size and age composition. The age-specific prevalence component includes any changes embedded in the modeled rates; calling it an epidemiological component does not isolate biological risk, diagnosis or policy. Clinical guidelines support individualized reproductive and broader health assessment Teede et al. (2023); Becker et al. (2022); these ecological comparisons cannot select interventions, estimate treatment benefits or demonstrate service inequity. They can identify patterns that surveillance and independent studies should examine.

The independent United Nations population audit offers a targeted check of the study's central *measurement and comparison* mechanism. Despite age-specific population differences between the UN WPP 2024 and GBD 2023 series, the composition contribution to the modeled PCOS China–reference contrast was +86.76 rather than +88.65 per 100,000, and its share of the net change was 57.6% rather than 58.5%. That agreement increases confidence that the substantive benchmark-composition result is not an artifact of one demographic population series. However, UN WPP is not an independent dataset of infertility outcomes: it supplies population weights while the GBD disease rates are held fixed. Related census inputs may also be shared between demographic models. Neither the apparent numerical stability nor a stable sign establishes a clinically meaningful change, and we have not recovered the missing jointly indexed GBD

posterior draws.

The changing country mix is not merely a technical detail. Within this selected 18-country comparator, UN WPP estimates suggest India's age-weighted reference share rose by about 10.74 percentage points between 1990 and 2023, whereas Japan and the United States lost share. Thus the pooled reference is a changing demographic estimand even after fixed age standardization. Reporting fixed-1990 and equal-country alternatives exposes which descriptive interpretations depend on policy choices about the comparator. These alternatives should not be used selectively to manufacture significance or to imply that a historical fixed mix is inherently the correct benchmark. The near-zero 2023 endometriosis contrast remains endpoint-sensitive despite the UN audit, reinforcing the need for independent diagnostic-series evidence rather than a claim of clinical convergence.

The principal GBD limitation remains unavailable joint uncertainty draws. We cannot propagate GBD uncertainty through pooling, standardization, subtype shares or the new decomposition. A 58.5% or 69.4% point estimate should therefore not be read as a precisely established population parameter. Country exclusions test dependence on the reference; they cannot replace draw-based intervals. Bootstrap resampling of the deliberately selected G20 countries would answer another question and would not recover GBD uncertainty. Conditional trend intervals are confined to the supplement and do not establish cross-country differences.

Other limitations include sparse or heterogeneous upstream evidence, modeled cause allocation, the distinction between impairment and disease prevalence, the economically selected comparison group, and the absence of definitionally matched external validation of GBD. Standard weights were available only as a rounded published reproduction. Several analyses were added after inspection, including endpoint checks and the reference-composition decomposition; their exploratory status requires transparent reporting and future corroboration. No release-to-release comparison was performed, so differences from previous publications cannot be assigned to real changes rather than GBD revision. These constraints limit the strength of inference even after formatting and numerical checks.

5 Conclusions

Within GBD 2023 point estimates, China's apparent convergence toward other G20 countries for PCOS-related infertility partly reflected changing reference-country population shares and concealed contrasting primary and secondary infertility patterns. Endometriosis near parity in 2023 depended strongly on the terminal year. International summaries should distinguish subtype, age and reference composition; these results require propagated uncertainty and independent corroboration before causal or policy conclusions.

List of abbreviations

AAPC, average annual percentage change; ARIMA, autoregressive integrated moving average; ASPR, age-standardized prevalence rate restricted here to ages 15–49; GATHER, Guidelines for Accurate and Transparent Health Estimates Reporting; GBD, Global Burden of Disease; G20, Group of Twenty; IHME, Institute for Health Metrics and Evaluation; MAE, mean absolute error; OLS, ordinary least squares; PCOS, polycystic ovarian syndrome; YLD, years lived with disability.

Authors' roles

XX: Author 1—conceptualization, data curation, formal analysis, writing; XX: Author 2—validation, interpretation and critical revision; XX: Corresponding author—supervision, verification, critical revision. Replace these illustrative roles with accurate author-specific contributions. Each author must approve the final manuscript and sign the journal authorship form before submission.

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Disclosures

XX: author-specific financial and non-financial interests, including applicable relationships within the preceding 36 months, must be confirmed; do not interpret this placeholder as a declaration of no conflicts.

Availability of data and materials

The independently sourced United Nations WPP 2024 female population table is publicly available from the UN Population Division. The 19-country age–year subset, original-file SHA-256, full crosswalk and analytical source code are provided in the accompanying reproducibility package. The UN data were used for reference-country weights only; original GBD clinical/infertility estimates are governed by IHME conditions.

The derived datasets supporting the reported results and their analysis code are provided in Additional file 2 and the public GitHub repository PCOS research workspace (2026). Original GBD inputs are obtained from IHME through the Results Tool IHME (2026), subject to its data-use terms and access requirements. Exact dimensions, identifiers, download date, hashes and acquisition specifications are supplied. The original CSVs are retained in the research archive; they are not included in the public additional file. Re-running the full source-to-result pipeline requires obtaining these IHME inputs. The public repository contains the analysis revision and derived tables. XX: add a persistent archival DOI or other journal-accepted data/code identifier if available before submission, and verify original source-data licence terms.

Ethics approval and consent to participate

Only aggregate GBD model outputs and UN demographic summaries were analysed in the present study, without direct access to identifiable participants. XX: the responsible institu-

tion must confirm whether ethics approval, exemption or a non-human-subject determination applies; record the exact determination and identifier if required.

Consent for publication

Not applicable: the manuscript contains no individual person's data, images or identifiable details.

Additional files

Additional file 1 (PDF, with editable TeX source). Supplementary methods, secondary case-count and trend analyses, endpoint checks, forecast validation, acquisition audit and GATHER reporting map; supplementary tables and figures.

Additional file 2 (ZIP). Machine-readable derived result tables, executable Python source, configuration, dependencies, acquisition and validation manifests, and a README specifying licensed IHME inputs needed to reproduce the analysis. No synthetic research observations or licensed raw GBD files are included. The separate NSFG study is not part of this manuscript or its additional files.

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Figure legends

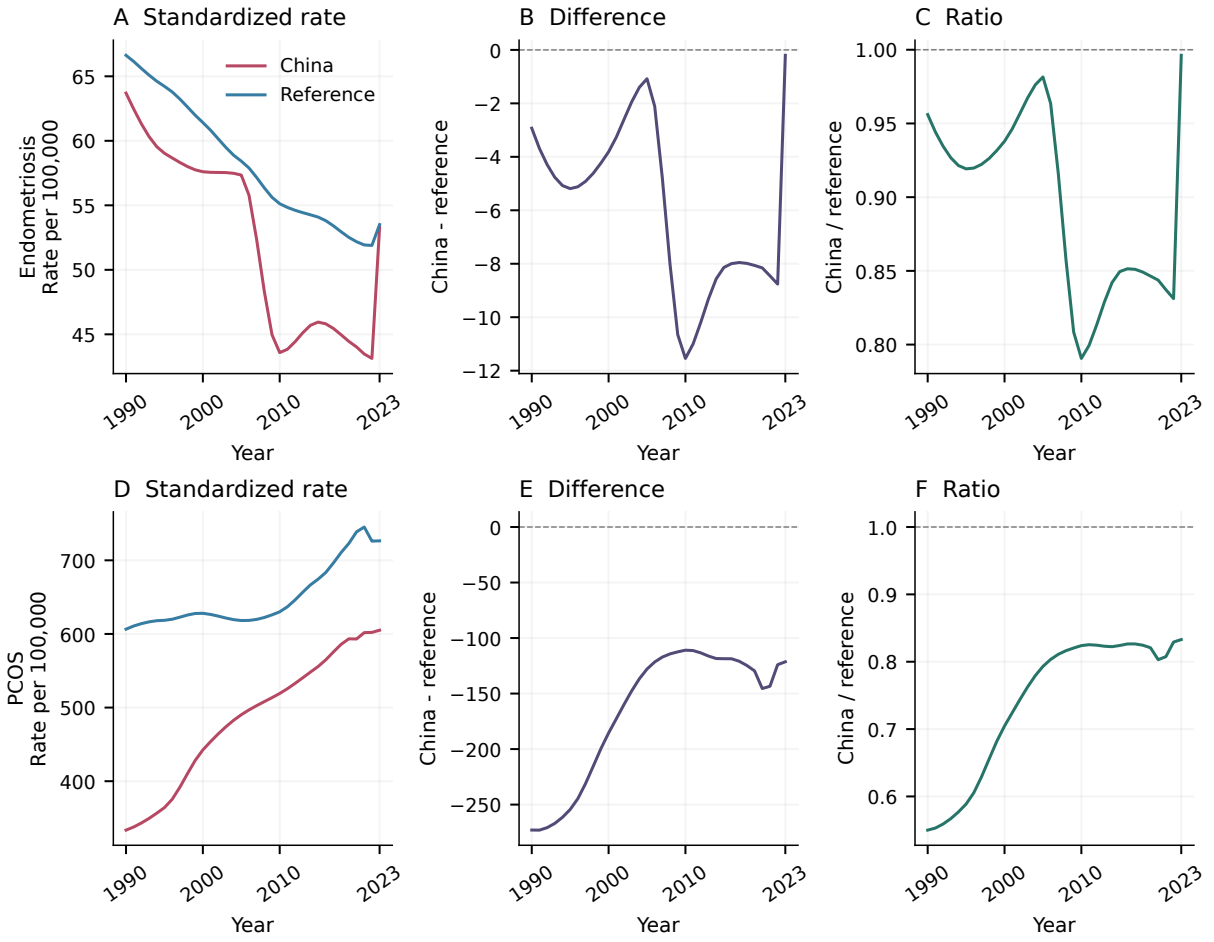


Figure 1: Annual total infertility rates and China–reference contrasts. Rates are standardized within ages 15–49. The reference pools population and modeled cases within age across 18 countries. Point trajectories have no propagated GBD uncertainty. The terminal-year endometriosis change is a source-estimate feature requiring independent corroboration.

Alt text: Annual rates and China-minus-reference contrasts. PCOS rates rise while the negative contrast narrows. Endometriosis is endpoint-sensitive.

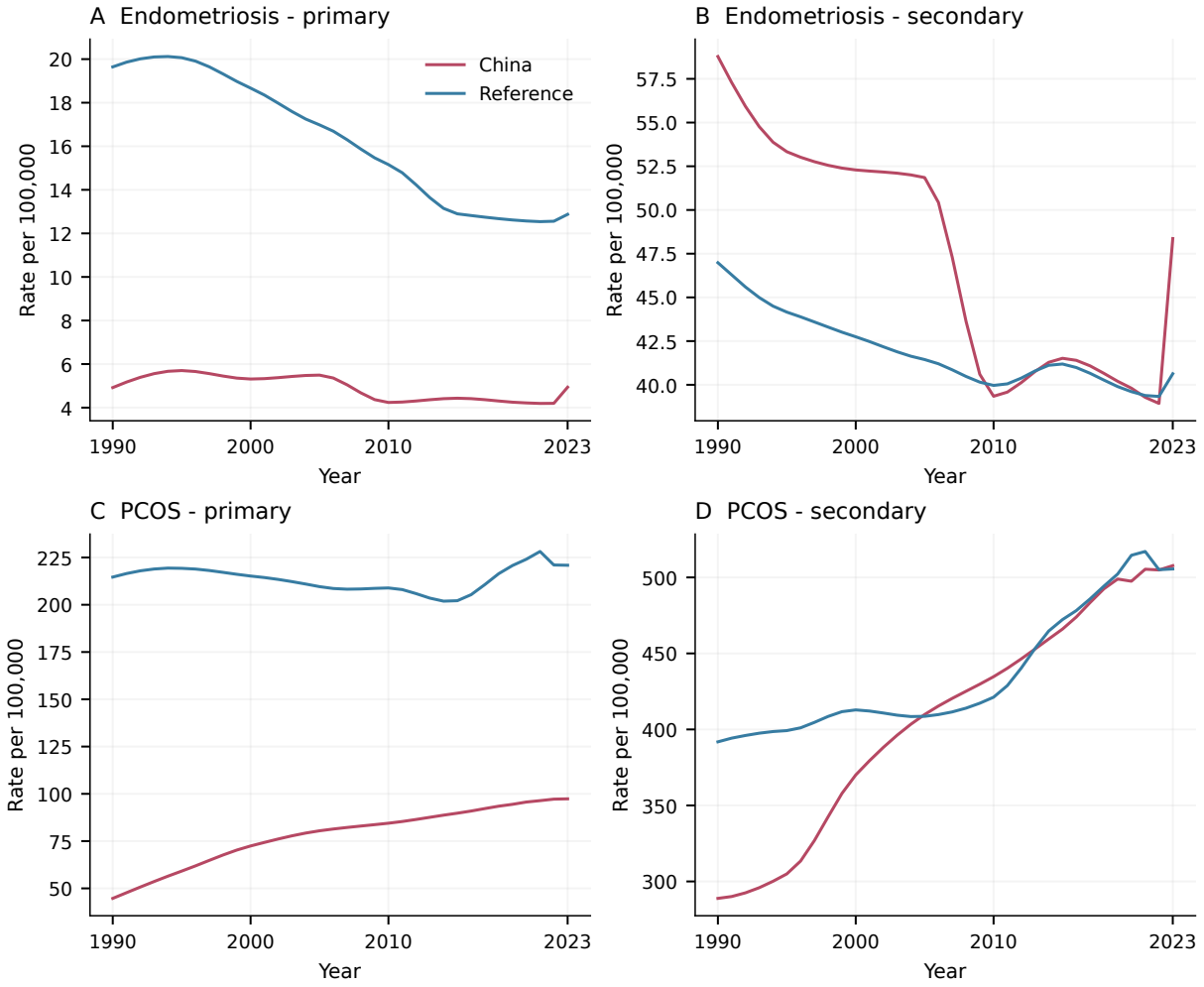


Figure 2: Primary and secondary infertility trajectories. The same restricted-age standard and annual population-pooled reference are used for both types. Opposing subtype contrasts can cancel in the total. These modeled categories do not track transitions in individual women.

Alt text: Primary and secondary trajectories differ and can cancel in the total contrast.

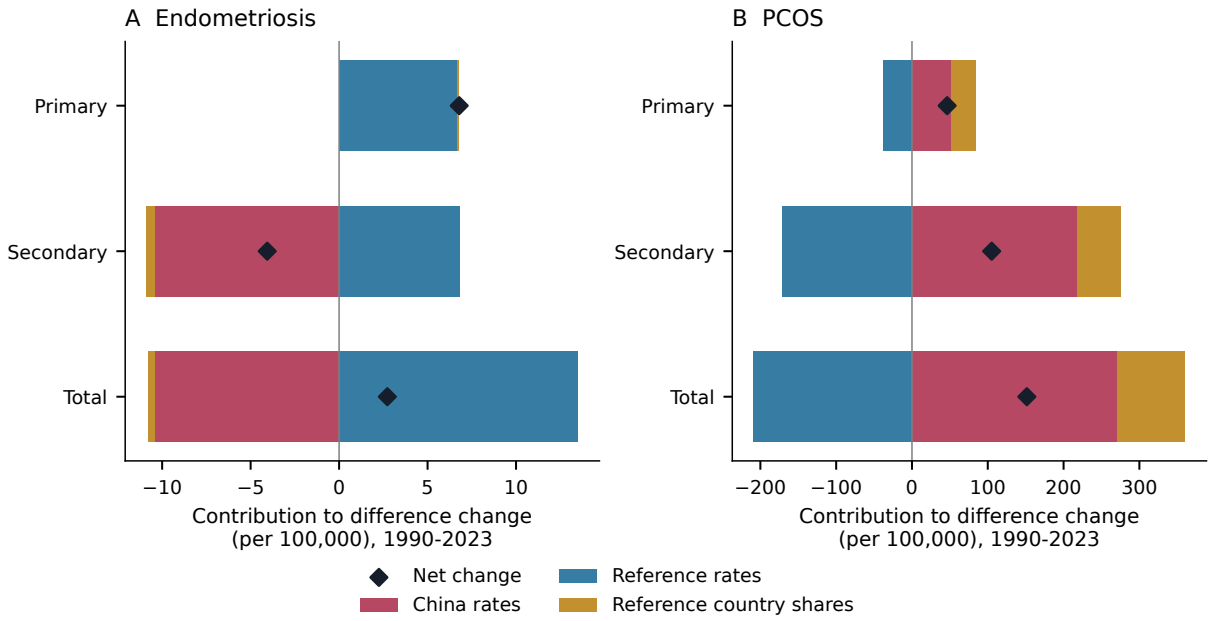


Figure 3: Rate and reference-composition contributions to contrast changes. Contributions sum to the 1990–2023 change in China minus reference, separately by infertility type. Diamonds denote net changes. Positive components move the signed contrast upward. Reference-country shares vary within age; fixed standard age weights are used throughout. The decomposition was added during manuscript review and is descriptive.

Alt text: Signed rate and country-share components sum to contrast changes. Country composition contributes materially to PCOS.

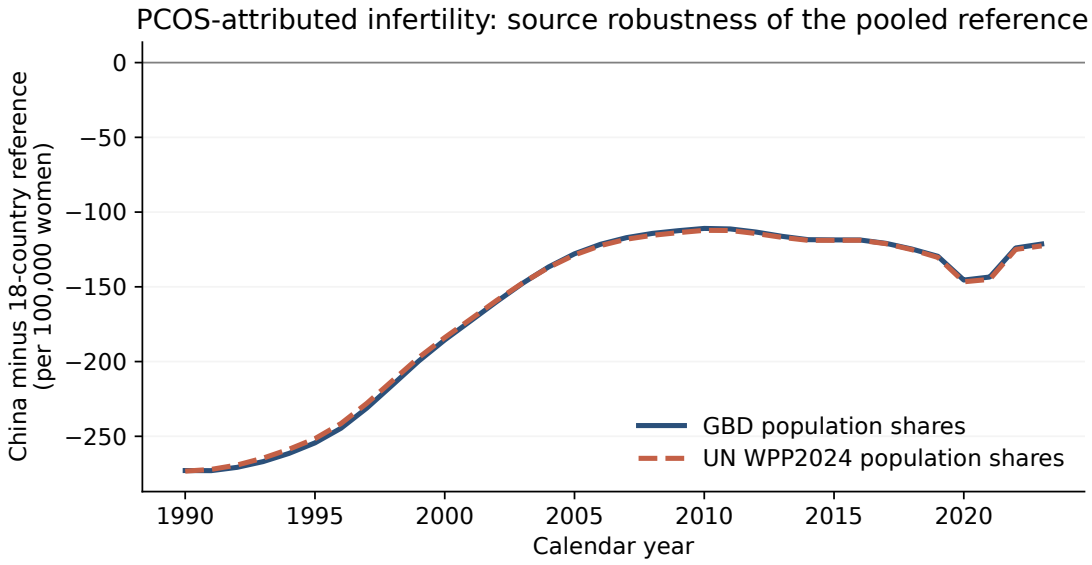


Figure 4: GBD and UN demographic-source sensitivity of the China-minus-reference PCOS contrast. Both scenarios hold GBD 2023 country- and age-specific disease rates constant; only the reference-country female population shares differ within age and year. Values are model-conditioned point estimates, not confidence intervals.

Alt text: Two closely similar annual curves of China minus 18-country PCOS infertility burden; GBD and UN population weighting produce similar results.

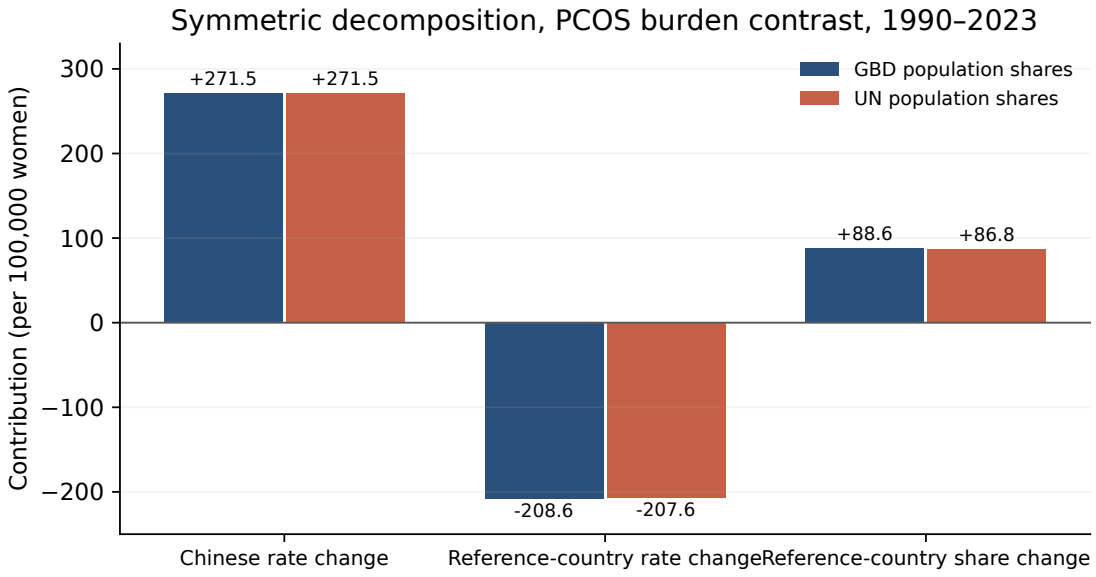


Figure 5: Population-source comparison of symmetric decomposition components. Chinese age-specific modeled prevalence rates, reference-country rates and changing reference shares contribute to the signed 1990–2023 China-minus-reference contrast; the three components sum exactly to the model-conditioned net change. UN shares substitute only the reference population composition, not GBD disease rates.

Alt text: Population composition is a substantial positive contribution to a narrowing modeled PCOS contrast under either GBD or UN population weights.